DAI-037 — Slowly Changing Customer Dimension (SCD Type 2)

Design / Decision Note

Goal. Preserve customer attribute history from change files, make reruns safe, and generate an AI-readable change report.

Architecture

CSV snapshots → validation → SHA-256 batch identity → transactional Python ingestion → SQLite dimension/history tables → JSON + Markdown reports.

SCD Type 2 rules

- First observation inserts the initial current version.
- Unchanged tracked attributes do not create a new version.
- A changed row closes the prior version at the incoming effective date and inserts a new current version.
- effective_to is exclusive; NULL indicates the current version.

Key decisions

SQLite and Python's standard library keep the assessment reproducible without external services. A SHA-256 hash of normalized content skips exact reruns. Database transactions avoid partial batches. The pipeline rejects out-of-order effective dates and duplicate customer IDs within a file rather than silently corrupting history. No external AI API is called; the deterministic JSON schema provides a clean integration point for a future AI assistant.

Validation and deliverables

Tests cover initial insert, history preservation after an update, idempotent reruns, unchanged attributes, out-of-order changes, duplicate keys, and report generation. Deliverables include runnable Python, SQL schema, synthetic CSVs, tests, README, and a 5–8 minute demo script.